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ENGS 93 Homework - Part 1

Descriptive Statistics and Normal Probability Distributions Dataset: Engineering Salaries 2021

This analysis introduces the dataset, summarizes its variables, examines the distribution of engineering salaries, and estimates probabilities using a fitted Normal distribution.

```
clear;
clc;
close all;
```

Question 1: Importing and Summarizing the Dataset

The Engineering Salaries 2021 dataset contains 10,000 observations and 18 variables. Each observation represents an engineering salary record with information about compensation, professional experience, employment characteristics, and demographic attributes.

The dataset contains both numerical and categorical variables, satisfying the requirements of the assignment. We import the data as a MATLAB table, preserving the original column headers, and examine the observed range or categories of every variable.

```
% the repository root is identified from the current matlab folder.
% this avoids temporary paths created during report publishing.
project_root = pwd;

% the parent directory is used when matlab starts inside part1.
if strcmp(string(project_root), string(fullfile( ...
        '/Users/ritisingh/Desktop/engs93_homework', 'part1')))
    project_root = fileparts(project_root);
end
```

```
% the dataset is stored in a shared folder at the repository root.
data_path = fullfile(project_root, 'data', ...
    'Engineering_Salaries_2021_10000.csv');

% this check prevents analysis from continuing with a missing dataset.
assert(isfile(data_path), ...
    'The salary dataset was not found in the data directory.');

% readtable preserves the csv column headers as table variable names.
data = readtable(data_path);

% matlab's summary function gives an overview of the dataset.
disp('Question 1: Dataset Summary');
summary(data);

% the table dimensions correspond to observations and variables.
num_observations = height(data);
num_variables = width(data);

fprintf('\nNumber of observations: %d\n', num_observations);
fprintf('Number of variables: %d\n', num_variables);
```

```
Question 1: Dataset Summary

data: 10000×18 table

Variables:

    source_record_id: double
    submission_date: datetime
    company: cell array of character vectors
    engineering_role: cell array of character vectors
    company_job_level: cell array of character vectors
    specialization: cell array of character vectors
    location: cell array of character vectors
    education: cell array of character vectors
    gender: cell array of character vectors
    base_salary: double
    annual_bonus: double
    annual_stock_value: double
    total_annual_compensation: double
    years_of_experience: double
    years_at_company: double
    company_sample_count: double
    role_sample_count: double
    company_role_sample_count: double

Statistics for applicable variables:

                              NumMissing          Min
Median              Max                Mean              Std

    source_record_id                  0                52794
67839               83874          6.8103e+04          8.8634e+03
```

```
    submission_date                  0           2021-01-01
2021-04-29        2021-08-17        2021-04-30        1560:05:19
    company
0

    engineering_role
0

    company_job_level
0

    specialization
0

    location
0

    education
0

    gender
0

    base_salary                      0                4000
136000             893000        1.3557e+05        6.0185e+04
    annual_bonus                     0                   0
12000             420000        1.7548e+04        2.3225e+04
    annual_stock_value               0                   0
21000             900000        5.0138e+04        8.1458e+04
    total_annual_compensation        0                10000
175000            2372000            203625        1.3707e+05
    years_of_experience              0                   0
5                 40             6.7062            5.5657
    years_at_company                 0                   0
2                 40             2.5620            3.2057
    company_sample_count             0                   1
47               1077           229.1014          340.5069
    role_sample_count                0                 121
8675               8675        7.6009e+03        2.7491e+03
    company_role_sample_count        0                   1
29                930           175.5182          281.9470


Number of observations: 10000
Number of variables: 18
```

Question 1: Variable Ranges and Categories

For numerical variables, the observed range is defined by the minimum and maximum values. Categorical variables do not have meaningful numerical ranges, so their distinct categories are reported instead.

Submission dates are handled separately because they represent a chronological interval rather than a continuous measurement.

```
% the original variable names allow each column to be inspected
% without manually specifying all eighteen variables.
variable_names = data.Properties.VariableNames;

for i = 1:num_variables

    variable_name = variable_names{i};
    column = data.(variable_name);

    fprintf('\n%s:\n', variable_name);

    if strcmp(variable_name, 'submission_date')

        % the date column may be imported as text or as datetime.
        if ~isdatetime(column)
            dates = datetime(string(column), ...
                'InputFormat', 'yyyy-MM-dd');
        else
            dates = column;
        end

        fprintf('  earliest date: %s\n', ...
            char(string(min(dates))));
        fprintf('  latest date: %s\n', ...
            char(string(max(dates))));

    elseif isnumeric(column)

        % only finite observations contribute to numerical ranges.
        valid_values = column(isfinite(column));

        if ~isempty(valid_values)
            fprintf('  minimum: %.2f\n', min(valid_values));
            fprintf('  maximum: %.2f\n', max(valid_values));
        else
            fprintf('  no valid numerical observations\n');
        end

    else

        % unique labels describe the categories present in the data.
        category_names = unique(string(column));
        category_names = category_names( ...
            ~ismissing(category_names));

        fprintf('  number of categories: %d\n', ...
            numel(category_names));

        % a short preview keeps the published report readable.
        preview_count = min(numel(category_names), 10);

        for j = 1:preview_count
            fprintf('    %s\n', char(category_names(j)));
        end
```

```
        if numel(category_names) > preview_count
            fprintf('    %d additional categories\n', ...
                numel(category_names) - preview_count);
        end

    end
end
```

```
source_record_id:
  minimum: 52794.00
  maximum: 83874.00

submission_date:
  earliest date: 2021-01-01
  latest date: 2021-08-17

company:
  number of categories: 953
    10x Genomics
    23andMe
    2U
    3M
    7-Eleven
    8x8
    ABB
    ADP
    AIG
    AMD
    943 additional categories

engineering_role:
  number of categories: 4
    Hardware Engineer
    Mechanical Engineer
    Software Engineer
    Software Engineering Manager

company_job_level:
  number of categories: 959
    0
    006
    1
    1-2
    10
    107
    109
    10R
    11
    12
    949 additional categories

specialization:
```

```
  number of categories: 461
    --
    3D Artist
    5G
    ADAS
    AEM
    API Development (Back-End)
    AR
    AR/VR
    ARVR
    ASIC
    451 additional categories

location:
  number of categories: 569
    Aachen, NW, Germany
    Aarhus, AR, Denmark
    Aberdeen Proving Ground, MD
    Albany, NY
    Albuquerque, NM
    Aliso Viejo, CA
    Allen Park, MI
    Alpharetta, GA
    American Fork, UT
    Ames, IA
    559 additional categories

education:
  number of categories: 6
    Bachelor's Degree
    Highschool
    Master's Degree
    Not reported
    PhD
    Some College

gender:
  number of categories: 4
    Female
    Male
    Not reported
    Other

base_salary:
  minimum: 4000.00
  maximum: 893000.00

annual_bonus:
  minimum: 0.00
  maximum: 420000.00

annual_stock_value:
  minimum: 0.00
  maximum: 900000.00
```

```
total_annual_compensation:
  minimum: 10000.00
  maximum: 2372000.00

years_of_experience:
  minimum: 0.00
  maximum: 40.00

years_at_company:
  minimum: 0.00
  maximum: 40.00

company_sample_count:
  minimum: 1.00
  maximum: 1077.00

role_sample_count:
  minimum: 121.00
  maximum: 8675.00

company_role_sample_count:
  minimum: 1.00
  maximum: 930.00
```

Question 1: Interpretation

The dataset contains 10,000 observations and 18 variables, meeting the assignment's requirements for sample size and variable diversity.

The numerical variables include base salary, annual bonus, annual stock value, total annual compensation, years of professional experience, and years at the current company. The dataset also includes identifiers and sample-frequency counts, which are numerical but should not be treated as continuous measurements.

Base salary ranges from $4,000 to $893,000, and total annual compensation ranges from $10,000 to $2,372,000. Annual bonuses range from $0 to $420,000, while annual stock values range from $0 to $900,000.

Years of professional experience and years at the current company both range from 0 to 40 years. The submission dates extend from January 1, 2021, through August 17, 2021.

The categorical variables include company, engineering role, company job level, specialization, location, education, and gender. These variables describe differences among the salary records rather than quantities with minimum and maximum numerical values.

The observed ranges indicate considerable variation in compensation and professional experience. This variation makes the dataset useful for investigating distributions and applying statistical methods.

Question 2: Descriptive Statistics

We selected base salary and years of professional experience for our descriptive statistical analysis.

Base salary represents reported annual salary excluding additional compensation such as bonuses and stock awards. Years of experience measures the reported professional tenure of an engineer.

For each variable, we estimate the mean and sample standard deviation. The mean describes central tendency, while the standard deviation describes the typical spread around the mean.

```
% the two selected variables are stored separately for clarity.
salary = data.base_salary;
experience = data.years_of_experience;

% non-finite observations are excluded from the calculations.
salary = salary(isfinite(salary));
experience = experience(isfinite(experience));

% the mean summarizes the central value of each numerical variable.
salary_mean = mean(salary);
experience_mean = mean(experience);

% matlab's default std function uses n-1 in the denominator,
% producing the usual sample standard deviation.
salary_std = std(salary);
experience_std = std(experience);

fprintf('\nQuestion 2: Descriptive Statistics\n');

fprintf('\nBase salary:\n');
fprintf('  mean: $%.2f\n', salary_mean);
fprintf('  standard deviation: $%.2f\n', salary_std);

fprintf('\nYears of experience:\n');
fprintf('  mean: %.4f years\n', experience_mean);
fprintf('  standard deviation: %.4f years\n', experience_std);
```

```
Question 2: Descriptive Statistics

Base salary:
  mean: $135574.20
  standard deviation: $60185.29

Years of experience:
  mean: 6.7062 years
  standard deviation: 5.5657 years
```

Question 2: Interpretation of Descriptive Statistics

The average base salary in the dataset is $135,574.20, with a sample standard deviation of $60,185.29. This relatively large standard deviation indicates substantial variability in base compensation among the engineering salary records.

The average years of professional experience is 6.7062 years, with a sample standard deviation of 5.5657 years. The substantial variation in experience suggests that the dataset includes individuals at different stages of their engineering careers.

These statistics describe the center and variability of the observed data. However, the mean and standard deviation alone do not indicate whether a variable is Normally distributed. We examine the distribution graphically in the next section.

Question 2: Normalized Histogram and Normal Density

We examine the distribution of annual base salary using a probability-density-normalized histogram.

Unlike a frequency histogram, a density-normalized histogram is scaled so that its total area equals one. This allows a direct comparison with a continuous probability density function.

We construct a Normal distribution using the observed sample mean and sample standard deviation. A close match between the histogram and the fitted curve would suggest approximate Normality.

```
% a finely spaced salary vector produces a smooth density curve.
x = linspace(min(salary), max(salary), 1000);

% the normal density is evaluated using the estimated sample
% parameters rather than assuming a theoretical mean or variance.
normal_pdf = (1 / (salary_std * sqrt(2*pi))) .* ...
    exp(-0.5 * ((x - salary_mean) ./ salary_std).^2);

% the histogram is normalized to probability density so its
% scale matches the fitted normal density.
fig_salary = figure('Color', 'w');

histogram(salary, 40, ...
    'Normalization', 'pdf', ...
    'FaceAlpha', 0.6);

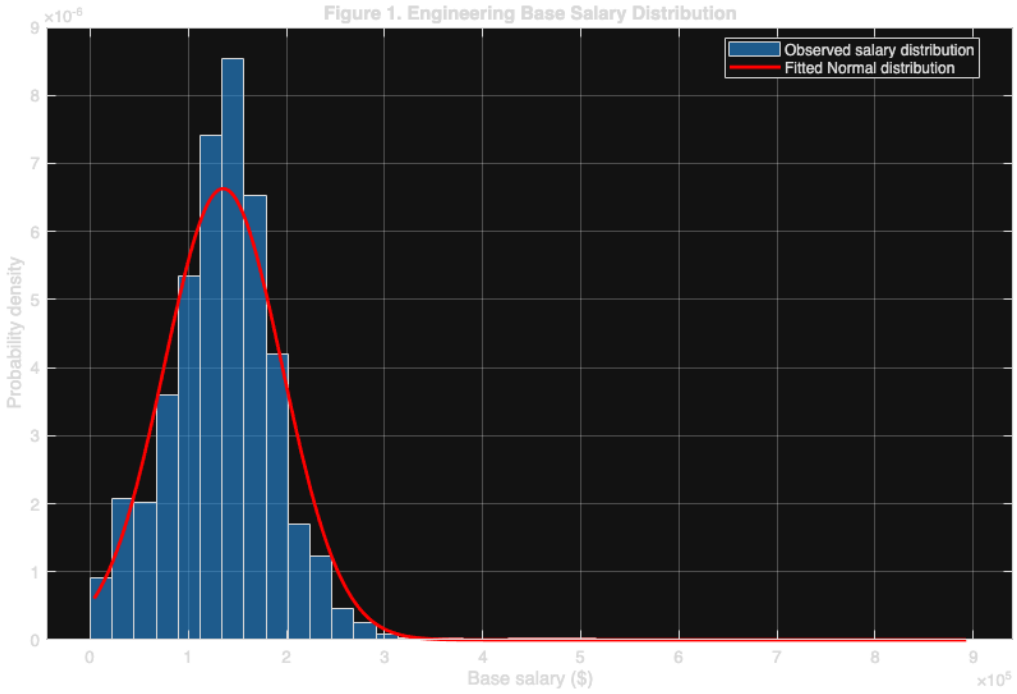
hold on;

% the red curve represents the fitted normal distribution.
plot(x, normal_pdf, 'r-', 'LineWidth', 2);

xlabel('Base salary ($)');
ylabel('Probability density');
title('Figure 1. Engineering Base Salary Distribution');

legend('Observed salary distribution', ...
    'Fitted Normal distribution', ...
    'Location', 'best');

grid on;
hold off;
```



Question 2: Assessment of Normality

The observed salary distribution is not approximately symmetric and bell-shaped. Instead, it exhibits positive skewness, with higher salary observations extending the right tail.

The Normal density curve provides a reasonable approximation near the central portion of the distribution, but it does not fully capture the asymmetry or the unusually high salaries.

Therefore, annual base salary is not well described by an exact Normal distribution. The fitted Normal model can still be used as an approximation, but the limitations of this assumption should be considered when interpreting probability estimates, especially in the tails.

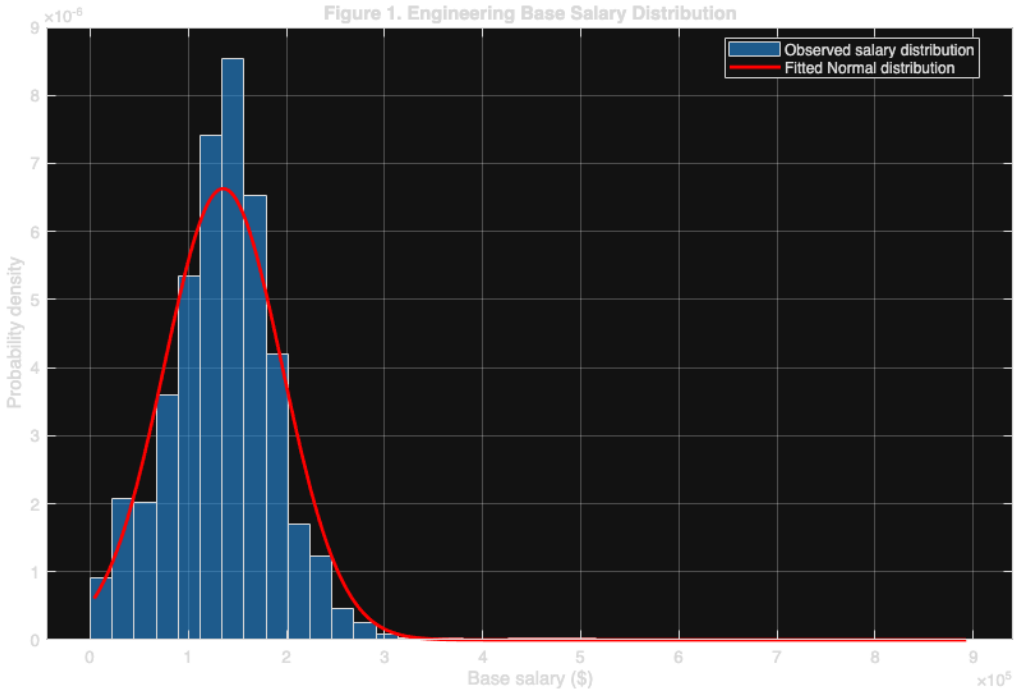
Question 2: Figure Export

The completed histogram is saved separately so that it can be included in the team's final report or reviewed outside MATLAB.

```
% the output path is shared across the project.
results_dir = fullfile(project_root, 'results', 'figures');

if ~isfolder(results_dir)
    mkdir(results_dir);
end

saveas(fig_salary, ...
    fullfile(results_dir, 'part1_salary_histogram.png'));
```



Question 3: Normal Probability Calculations

We use the fitted Normal distribution to estimate the proportion of engineering salaries below a selected threshold of $150,000.

The threshold is standardized using the observed salary mean and standard deviation. The resulting z-score indicates how many standard deviations the selected salary lies above the mean.

The cumulative Normal distribution then gives the probability that a salary is below the selected threshold.

```
% the salary threshold is an illustrative value within the
% observed range of the dataset.
salary_threshold = 150000;

% the z-score expresses the threshold relative to the mean
% and standard deviation of the fitted normal distribution.
z_score = (salary_threshold - salary_mean) / salary_std;

% the error function evaluates the normal cumulative probability
% without requiring the statistics and machine learning toolbox.
prob_less = 0.5 * (1 + erf(z_score / sqrt(2)));

% the complement gives the probability above the threshold.
prob_greater = 1 - prob_less;

% the empirical proportion provides a comparison against the
% observed dataset rather than the fitted normal model.
observed_proportion = mean(salary < salary_threshold);
```

```
fprintf('\nQuestion 3: Normal Probability\n');

fprintf('Salary threshold: $%.2f\n', salary_threshold);
fprintf('Z-score: %.4f\n', z_score);

fprintf('P(salary < $%.0f): %.4f\n', ...
    salary_threshold, prob_less);

fprintf('P(salary > $%.0f): %.4f\n', ...
    salary_threshold, prob_greater);

fprintf('Observed proportion below threshold: %.4f\n', ...
    observed_proportion);
```

```
Question 3: Normal Probability
Salary threshold: $150000.00
Z-score: 0.2397
P(salary < $150000): 0.5947
P(salary > $150000): 0.4053
Observed proportion below threshold: 0.5992
```

Question 3: Interpretation of Salary Probability

Using the fitted Normal distribution, the probability that an engineering base salary is below $150,000 is approximately 0.5947, or 59.47%.

The corresponding probability above $150,000 is approximately 0.4053, or 40.53%.

In the observed dataset, 59.92% of salary records fall below $150,000. The difference between the theoretical estimate and the observed proportion is approximately 0.45 percentage points.

This relatively small difference suggests that the fitted Normal model approximates the observed probability reasonably well at the selected threshold. However, agreement at one threshold does not imply that the entire distribution is Normally distributed.

In particular, the positive skewness observed in Question 2 indicates that the model may be less accurate for extreme salaries.

Question 3: Additional Probability for Experience

For comparison, we also estimate the probability that reported professional experience is below 10 years using a Normal distribution fitted to the experience variable.

This second calculation follows the same method as the salary calculation and demonstrates how the procedure applies to another numerical variable.

```
% the experience threshold is selected within the observed range.
experience_threshold = 10;

% the standardized experience value is relative to its own
% estimated mean and sample standard deviation.
```

```
experience_z = (experience_threshold - experience_mean) ...
    / experience_std;

% the fitted normal probability is evaluated at ten years.
experience_prob_less = 0.5 * ...
    (1 + erf(experience_z / sqrt(2)));

% the observed fraction provides an empirical comparison.
experience_observed = mean(experience < experience_threshold);

fprintf('\nAdditional experience probability:\n');

fprintf('P(experience < %.0f years): %.4f\n', ...
    experience_threshold, experience_prob_less);

fprintf('Observed proportion: %.4f\n', experience_observed);
```

```
Additional experience probability:
P(experience < 10 years): 0.7230
Observed proportion: 0.7418
```

Question 3: Interpretation of Experience Probability

The fitted Normal distribution estimates that approximately 72.30% of the observations have fewer than 10 years of professional experience.

The actual observed proportion is 74.18%. The model therefore underestimates the observed proportion by approximately 1.88 percentage points.

As with salary, this difference reflects the limitations of representing the empirical distribution with a fitted Normal model. The model-based probabilities are approximations rather than exact descriptions of the observed data.

Part 1: Conclusion

Part 1 established an initial statistical understanding of the Engineering Salaries 2021 dataset.

The descriptive statistics reveal substantial variation in annual base salary and professional experience. The histogram shows that base salary is positively skewed rather than approximately Normal.

Although the fitted Normal model does not reproduce the entire salary distribution, it provides a useful approximation for estimating probabilities at selected thresholds.

These findings provide a foundation for the confidence interval estimation and hypothesis testing conducted in Part 2.

```
fprintf('\nPart 1 analysis complete.\n');
```

```
Part 1 analysis complete.
```

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